Week 4 — Sampling & the CLT

Business Forecasting

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My code (.R) Notes

Where we are

Describing data → Reasoning under uncertainty → Regression → Time series forecasting

First week of reasoning under uncertainty. Your data is only a *sample*, so every number you computed so far is a guess. This week: how much those guesses move, and why they end up bell-shaped.

- This week (W4): population vs sample and parameter vs statistic, the random sample, the estimator and its properties (bias, variance, MSE), the sampling distribution, and the Central Limit Theorem.
- Next week (W5) turns this into confidence intervals.

Motivating example

You run a bunch of Airbnbs in Mexico City. Should you invest more in cleaning?

- Concretely: can you charge a higher price if your cleanliness score exceeds 4.5?
- Get a sample of listings and compare the price of
  - those with cleanliness score below 4.5 (dirty)
  - and above 4.5 (clean)

Motivating example: the sample

We have a sample of 200 listings (`price`, cleanliness `clean`):

id	review_scores_cleanliness	price	clean
40032982	3.00	1023	Dirty
21962322	4.50	4500	Dirty
41841538	4.50	380	Dirty
624813934659858771	3.40	1350	Dirty
47030021	4.29	684	Dirty
46736994	4.29	451	Dirty

Group	Listings	Mean price
Clean (score > 4.5)	100	1245
Dirty (score ≤ 4.5)	100	869

In these 200 listings, clean apartments charge 377 pesos more per night. Cleaning looks worth paying for.

But can you spend money on that number?

- Would a *different* 200 listings have given 377 — or 200, or 600?
- Is the gap across the *whole city* also 377, or could it be far smaller?
- How far from the truth can any number computed from 200 listings be?

How much of that can we answer with these 200 alone?

1 · Population, sample, parameter, statistic

The four words the rest of the course is written in.

Statistical language

- Population: the entire group you want to learn about. Usually too big to measure.
  - Every Airbnb listing in Mexico City.
- Parameter: one number describing the population. Fixed, unknown, never observed.
  - μ_c and μ_d — the mean price of every clean and every dirty apartment in the city. What you actually want is the gap $\mu_c - \mu_d$.
- Sample: the part of the population you actually have data on.
  - Your 200 listings — 100 clean, 100 dirty.
- Statistic: one number computed from the sample. You can always calculate it, and it lands somewhere different every sample.
  - $\bar{x}_c = 1{,}245$ and $\bar{x}_d = 869$, so the gap you measured is 377 pesos.

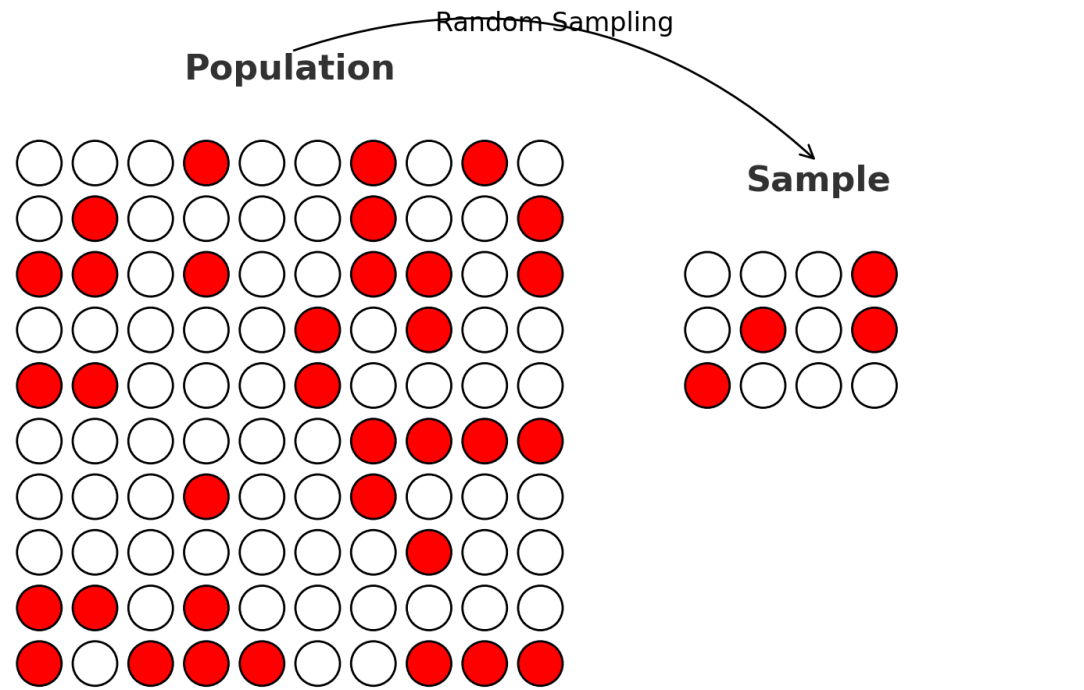
Parameter belongs to the population, statistic belongs to the sample. Greek letters (μ, σ, π) for parameters; Latin letters or hats ($\bar{x}$, s, $\hat{\pi}$) for statistics.

- A parameter does not have to be a mean. To know how many people in Mexico have diabetes, the population is everyone in Mexico and the parameter is a proportion:

$$\pi = \frac{D}{N}$$

→ More situations to sort out

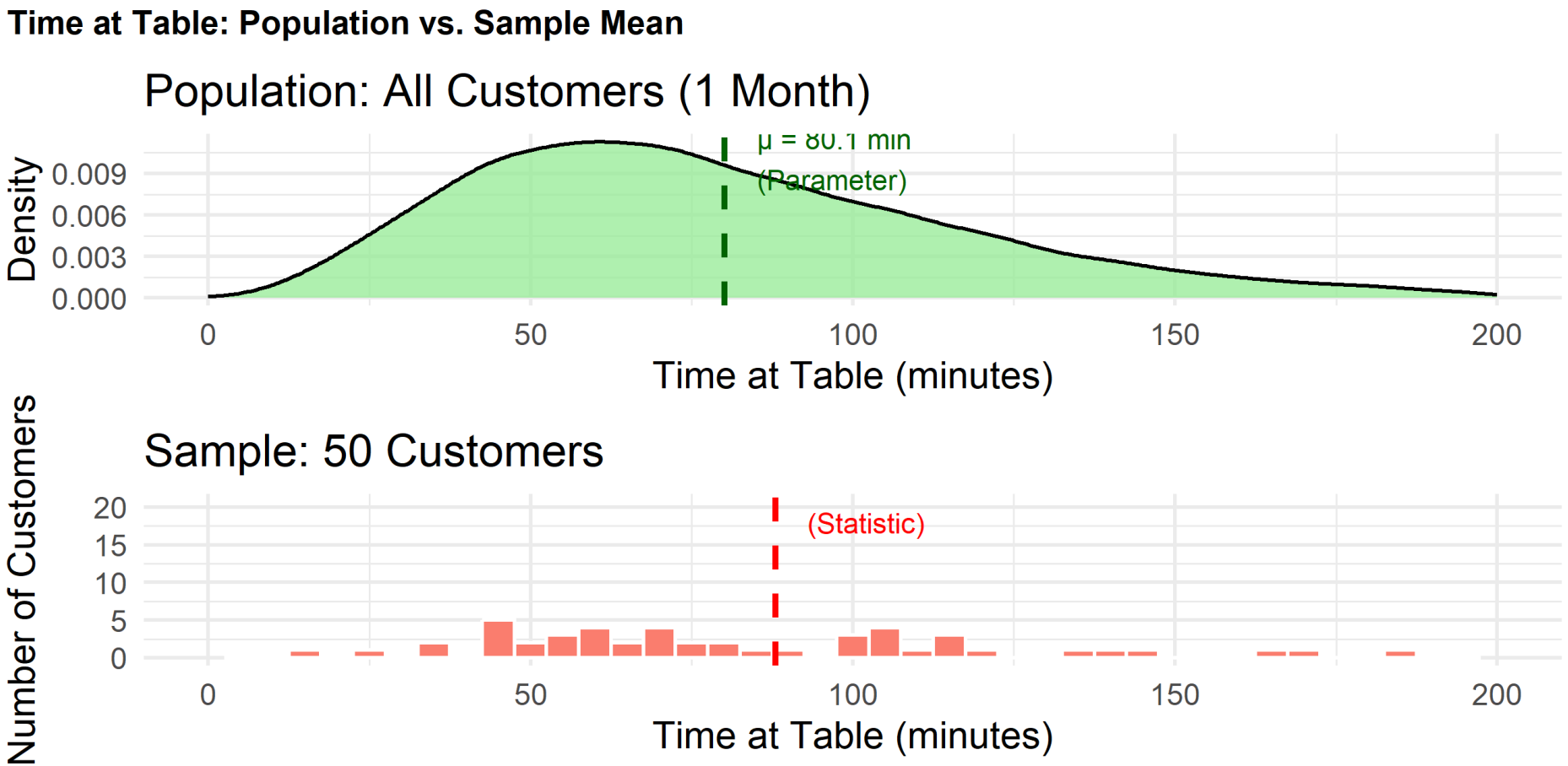
A parameter you cannot see, a statistic you can



Parameter (Population Diabetes %): 34.00% Statistic (Sample Diabetes %): 33.33%

- The parameter π belongs to the population on the left: fixed, unknown, never observed.
- The statistic $\hat{\pi}$ — the share of diabetics among the people you actually surveyed — belongs to the sample on the right. It is a guess at the parameter, computed from the sample.
- $\hat{\pi}$ is what you can calculate; π is what you wanted. Everything from here on is about the gap between the two.

The same picture for a continuous variable



- Top: every diner who sat down over a month, and the population mean μ — the parameter.
- Bottom: the 50 diners you actually timed, and the sample mean $\bar{x}$ — the statistic.
- Same shape, ragged instead of smooth, and the two dashed lines do not land in the same place. Draw a different 50 and the red line moves; the green one never does.

Why you almost never have the whole population

- Checking everything costs more than the answer is worth. *Decision:* should we pay to rebuild the billing system? *Parameter:* π, the share of all 200,000 invoices that contain an error. Checking all of them takes a team a month. Checking 300 takes an afternoon.

- Measuring destroys the product. *Decision:* how long a warranty can we offer? *Parameter:* μ, the mean lifetime in hours of every battery we make. You only learn a battery's lifetime by running it until it dies.

- Most people never answer. *Decision:* do we keep the new subscription plan? *Parameter:* μ, the mean satisfaction score of all 40,000 subscribers. You email all of them; 600 reply.

- The group has not happened yet. *Decision:* how much stock do we order for next month? *Parameter:* μ, mean daily demand next month. Next month's customers do not exist yet. All you have is the past.

Every one of these is a number about a group you cannot measure, decided from a group you can. The question is always the same: how far from the truth can that number be?

This week we cheat

In real work the parameter is unknown, so you can never check your answer.

Here we can. Those 200 listings were drawn from every Airbnb listing in Mexico City — and for this deck we have all of them:

```
population <- airbnb$price    # the nightly price of all 21,184 listings
```

So the parameters are no mystery. Across the whole city:

$$\mu_c = 1{,}195 \qquad \mu_d = 1{,}024 \qquad \mu_c - \mu_d = \mathbf{171} \text{ pesos}$$

Your 200 listings said the gap was 377. The truth is 171 — you would have valued cleaning at more than twice what it is worth. Nothing went wrong; that is simply what a sample of 200 does.

`mean(sample(population, 200))` is one estimate from a fresh scrape of 200 listings — we vary the sample size all week. Before you run anything: which of those two numbers changes if you repeat it?

Exercise 1 — The parameter and one estimate

1. Compute the parameter and one estimate from `n = 200`, and print them side by side.
2. Now produce five estimates, each from its own fresh sample of 200.
3. Report the range they cover.

Exercise | Start Over | Show Hint | Run Code

```
1  # Your code here
```

One is fixed, the other is not

- `mean(population)` — the parameter — is a single fixed number. It never moves.
- `mean(sample(population, 200))` — the estimate — is different on every draw.

The next four weeks are about that second line: how far does the estimate wander, and what can we still say about the truth after seeing only one draw of it?

Statistical language: sample & goal

- Sample: the part of the population we have data for.
  - Our 200 listings.
- Goal: what we want to learn about the population.
  - Is $\mu_c > \mu_d$? If so, by how much?
  - We don't know μ_c, μ_d — we guess them with an estimator and a random IID sample.

2 · Drawing the sample

What has to be true of the draw for the guess to mean anything.

What is a random sample?

- At random: each member of the population has an equal chance of being selected. Selecting is called *drawing* a sample.
- Random variable P_i: describes observation i *before* drawing — it could be any price from the distribution.
- Random sample: a collection of random variables $\{P_1, P_2, \ldots, P_n\}$.
- Observed value p_i: once realized, it is a constant from our sample — no longer random.

Before drawing the sample

The P_i are still random — no values yet:

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs						
Realized Values p_i						

Random sample — Draw 1

Drawing realizes each P_i into a concrete value p_i:

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	8451	9015	8161	9085	8268	1622
Realized Values p_i	120	150	800	200	1400	110

Random sample — Draw 2

A different draw from the same population gives different realized values:

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	1933	3947	3145	3773	6721	3373
Realized Values p_i	260	420	500	2120	800	1450

The point estimate changes every time we draw.

IID samples

- IID = Independent and Identically Distributed.
- Independent: selecting one unit P_i doesn't affect selecting another P_j.
- Identically distributed: all units P_i come from the same distribution.

3 · Estimators

The rule that turns a sample into a guess — and why it is random.

Estimators: intuition

- Our method of guessing the parameter from the data.
- A function of the random variables in the sample: $\hat{\theta} = f(P_1, \ldots, P_n)$.
- Because it is random, we can analyze its statistical properties.
- It cannot contain unknown quantities (like σ or μ).

Estimators: ones we've seen

- Sample mean: $\hat{\mu}_c = \bar{P} = \frac{\sum_n P_i}{n}$
- Sample standard deviation: $s_c = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(P_i - \bar{P})^2}$
- Each is a rule that turns a sample into a single guess.

Point estimate

- A single number computed from the realized data $\{p_1, \ldots, p_n\}$:

$$\bar{p} = f(p_1, \ldots, p_n) = \frac{\sum_n p_i}{n}$$

- No longer random — it is one number from one sample.

Example: the estimator — before drawing

You want the average nightly price in Mexico City. You will draw 8 listings at random and average them. Nothing has been drawn yet.

Random variable	P1	P2	P3	P4	P5	P6	P7	P8
Listing ID drawn	?	?	?	?	?	?	?	?
Realized price p	?	?	?	?	?	?	?	?

$$\hat{\mu} = \frac{P_1 + P_2 + \cdots + P_8}{8}$$

- Each P_i is the price of whichever listing turns up in slot i — a random variable, because the draw has not happened.
- So $\hat{\mu}$, built out of them, is a random variable as well. It has a distribution, a mean and a variance, before you see any data.

Example: the estimator — sample 1

Random variable	P1	P2	P3	P4	P5	P6	P7	P8
Listing ID drawn	4821	1330	7042	2914	6108	3567	9251	5480
Realized price p	120	150	800	200	1400	110	1900	800

$$\hat{\mu} = \frac{120 + 150 + 800 + 200 + 1400 + 110 + 1900 + 800}{8} = 685$$

One number, no longer random: the point estimate.

Example: the estimator — sample 2

Same city, same rule, eight *different* listings.

Random variable	P1	P2	P3	P4	P5	P6	P7	P8
Listing ID drawn	3145	3773	6721	3373	2102	5365	4453	3621
Realized price p	260	420	500	2120	800	1450	120	809

$$\hat{\mu} = \frac{260 + 420 + 500 + 2120 + 800 + 1450 + 120 + 809}{8} = 809.875$$

685 the first time, 809.9 the second. Nothing changed except which eight listings were drawn.

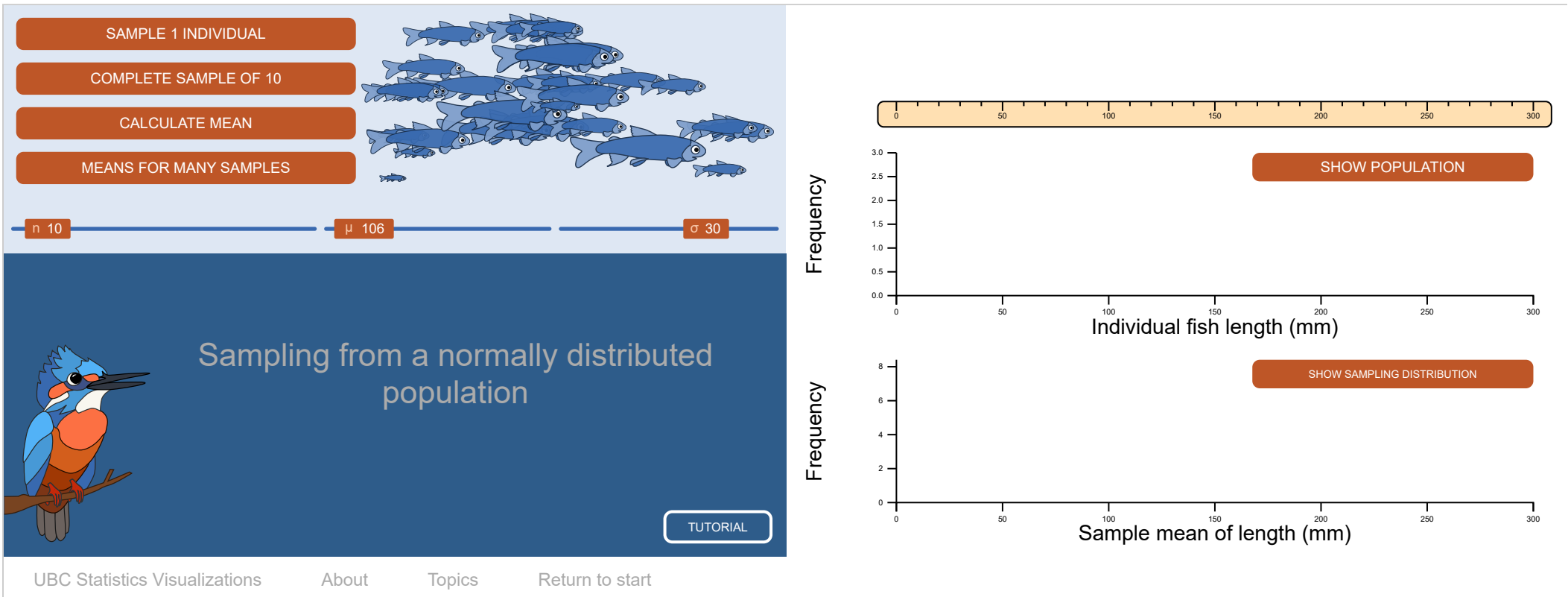
- The estimator $\hat{\mu}$ is the rule. It is random, and it has a distribution.
- The point estimate is one number that rule produced on one sample.
- Every question this week is about that distribution: where is it centred, and how wide is it?

One number isn’t enough

- The mean price of clean listings in our sample is $\bar{p}_c = 1245.4$ MXN — our point estimate.
- We can’t say how close this one number is to the true mean without knowing the population.
- But we can judge our *method* (the estimator) by looking at every estimate it could have produced.

Sampling distribution: explore it

Each estimator has its own sampling distribution — try a few:



4 · Judging an estimator

Bias, variance, and the trade-off between them.

What makes an estimator good?

The same parameter can be estimated in many different ways. For the average price you could take the sample mean, the median, the midpoint of the cheapest and dearest listing, or simply the first listing you happened to scrape. All of them are estimators.

So which ones are any good?

To judge a method rather than a single number, ask three things about its sampling distribution:

1. Where is it centred? — its expectation $E[\hat{\theta}]$. If that sits on the parameter, the estimator is unbiased: it is right *on average*.
2. How wide is it? — its variance $Var(\hat{\theta})$. How often does a single estimate land close to that centre?
3. What shape is it? — the whole distribution. Only the shape lets you attach confidence to a number: *"we are 95% confident the true mean is within 100 pesos of our estimate"*. That is next week's business.

Centre, spread, shape — we take them in that order.

The expectation of an estimator

Three symbols that are easy to confuse.

- θ — the parameter. One fixed, unknown number about the population: the average price of every listing in the city, or the clean-minus-dirty gap. It never moves, and you never see it.
- $\hat{\theta}$ — the estimator. The rule you apply to a sample. It is random: draw a different sample and it hands you a different number.
- $E[\hat{\theta}]$ — the expectation of the estimator. Take every sample you could have drawn, apply the rule to each, and average all those answers.

$E[\hat{\theta}]$ is not the average of your data. It is the centre of the sampling distribution — the average of numbers you never computed, from samples you never drew.

Bias: is the rule right on average?

$$Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- An estimator is unbiased when $E[\hat{\theta}] = \theta$ — for *any* value θ might take. Its sampling distribution is centred exactly on the parameter.
- That is a statement about the rule, not about your sample. An unbiased rule can still hand you a number far from θ; bias is about the centre, not the width.

- To see it: take many samples, compute the estimator in each, and average those values — does that average land on the parameter?
- In practice you have one sample, so you can never actually do that.
- But you can usually derive $E[\hat{\theta}]$ on paper, with no data at all. That is what the next few slides do.

Three candidate estimators

Same job: guess μ from an IID sample.

- $\hat{\theta}_1 = \frac{3X_1 + X_2}{4}$ — leans almost entirely on the first observation.
- $\hat{\theta}_2 = \frac{3X_1 + X_2}{5}$ — the same idea with different weights.
- $\hat{\theta}_3 = \bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$ — the sample mean, whatever n happens to be.

Which of them are unbiased? And is unbiased the same as good?

Example 1: $\hat{\theta}_1 = \frac{3X_1 + X_2}{4}$

- Every X_i has the same unknown mean $E(X_i) = \mu$.
- Expected value: $E(\hat{\theta}_1) = \frac{3}{4}\mu + \frac{1}{4}\mu = \mu$.
- Bias: $\mu - \mu = 0$ → unbiased. The weights add to 1, which is all that is needed.
- But it throws away X_3 and X_4 entirely. Is it a *good* estimator? (We come back to this.)

Example 2: $\hat{\theta}_2 = (3X_1 + X_2)/5$

- Same setup: $E(X_i) = \mu$.
- Expected value: $E(\hat{\theta}_2) = \frac{3}{5}\mu + \frac{1}{5}\mu = \frac{4}{5}\mu$.
- Bias: $\frac{4}{5}\mu - \mu = -\frac{1}{5}\mu$ → biased (weights don’t sum to 1).

Example 3: $\hat{\theta}_3 = \bar{X} = \frac{\sum X_i}{n}$ (the sample mean)

- Same setup: $E(X_i) = \mu$. Every observation gets the same weight $\frac{1}{n}$.
- Expected value: $E(\bar{X}) = \frac{\sum_n E(X_i)}{n} = \frac{n\mu}{n} = \mu$.
- Bias: 0 → unbiased. The sample mean is our workhorse estimator.
- Look at what that proof needed: only that each $E(X_i) = \mu$. We never had to know what μ is — and we never will.
- That is exactly why it is useful. Whatever the true average price turns out to be, the sample mean is aimed at it — so the number your one sample hands you is real information about μ.

Two unbiased estimators — is one of them better?

So we want an estimator that is unbiased. That rules out $\hat{\theta}_2$ — but it left *two* survivors, and it will often leave more than one. How do we choose between them?

The two survivors are $\hat{\theta}_1$ and $\hat{\theta}_3$. To compare them concretely, take a sample of four observations, so the sample mean is $\bar{X} = \frac{X_1+X_2+X_3+X_4}{4}$:

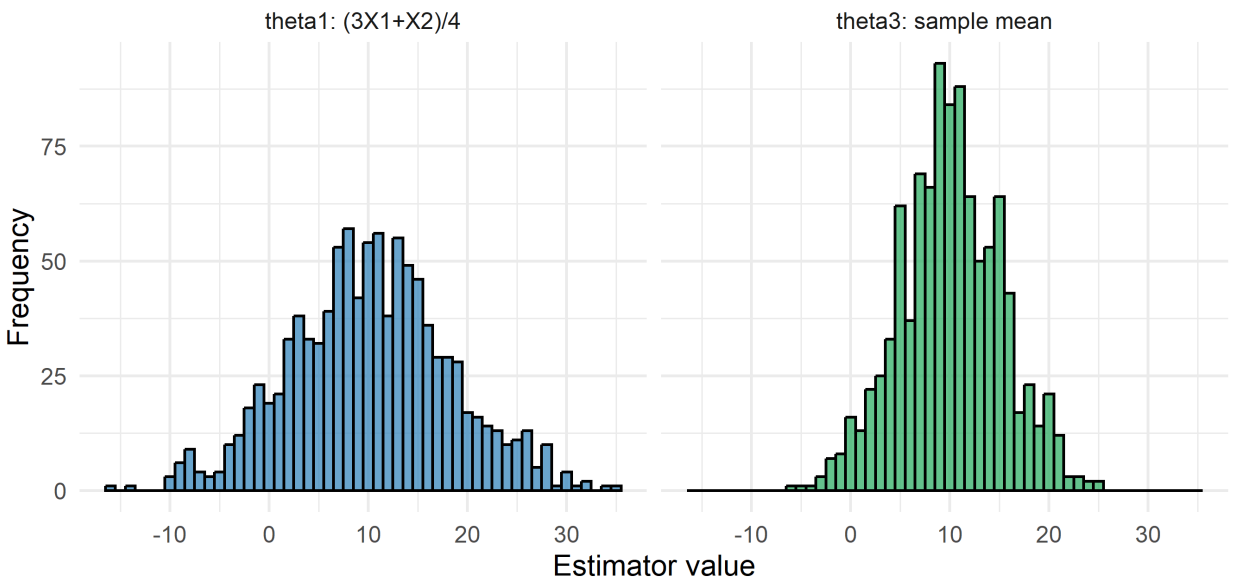
- $\hat{\theta}_1 = \frac{3X_1+X_2}{4}$ — three quarters of the weight on one observation.
- $\hat{\theta}_3 = \frac{X_1+X_2+X_3+X_4}{4}$ — the weight spread evenly over all four.

The weights sum to 1 in both, so both are unbiased. Bias cannot separate them.

Which one would you rather report — and what evidence would settle it?

Two unbiased estimators, 5,000 samples each

Each estimator computed in 5,000 samples of four draws from $X \sim \mathcal{N}(10, 10)$.



- Both histograms are centred on the true value — unbiased, exactly as the algebra said.
- But $\hat{\theta}_1$'s estimates are spread much wider. Plenty of them land 5 or more away from the truth; $\hat{\theta}_3$'s almost never do.
- You only ever draw one sample. A wide estimator is right *on average* and can still hand you an answer far from μ — and you would never know.
- So among unbiased estimators, the one whose estimates cluster tightest is the one you want. It is called the more efficient estimator.

To rank them we need a number for that spread — the variance of the estimator.

Variance of an estimator: the definition

$$Var(\hat{\theta}) = E\left[(\hat{\theta} - E[\hat{\theta}])^2\right]$$

Two rules turn that definition into numbers:

- a constant comes out squared: $Var(aX) = a^2 Var(X)$
- for independent draws the variance of a sum is the sum of the variances.

Apply them to each candidate and the numbers fall straight out.

Variance: worked examples

All three estimate the same μ from the same four draws. Only the weights differ.

Estimator	Weights	Variance	In units of σ^2
X_1 alone	$1,\ 0,\ 0,\ 0$	σ^2	1.00
$\hat{\theta}_1 = \dfrac{3X_1 + X_2}{4}$	$\frac{3}{4},\ \frac{1}{4},\ 0,\ 0$	$\dfrac{9\sigma^2 + \sigma^2}{16} = \dfrac{10\sigma^2}{16}$	0.63
$\hat{\theta}_3 = \bar{X}$	$\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}$	$\dfrac{\sigma^2}{4} = \dfrac{4\sigma^2}{16}$	$\mathbf{0.25}$

The more evenly the weight is spread, the smaller the variance. Spread it as evenly as possible and you get the sample mean:

$$Var(\bar{X}) = \frac{\sigma^2}{n}$$

It falls with every observation you add — which is exactly what the uneven weights give up.

→ Where σ^2/n comes from · → Where $10\sigma^2/16$ comes from

Relative efficiency

Among unbiased estimators, the smaller variance wins:

$$Var(\hat{\theta}_1) = \frac{10\sigma^2}{16} > \frac{4\sigma^2}{16} = Var(\hat{\theta}_3)$$

$$Eff(\hat{\theta}_3, \hat{\theta}_1) = \frac{Var(\hat{\theta}_1)}{Var(\hat{\theta}_3)} = \frac{10}{4} = 2.5$$

The sample mean is 2.5× as efficient.

Side note: what if variables aren't independent?

So far we assumed independence. Suppose X_1, X_2 are not independent (e.g. daily sales of two products in one store).

- $E(X_1 + X_2) = E(X_1) + E(X_2)$ (always true).
- $Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2\,Cov(X_1, X_2)$
- $Var(X_1 - X_2) = Var(X_1) + Var(X_2) - 2\,Cov(X_1, X_2)$

Mean Squared Error (MSE)

One number for how good an estimator is — how far it lands from the truth, on average:

$$MSE(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right]$$

It splits into exactly two pieces, and nothing is left over:

$$MSE(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2$$

- Variance — how much your answer moves when the sample changes.
- Bias^2 — how far the method is off on average.
- If the estimator is unbiased, the second term is zero and $MSE(\hat{\theta}) = Var(\hat{\theta})$.

→ The two-line derivation

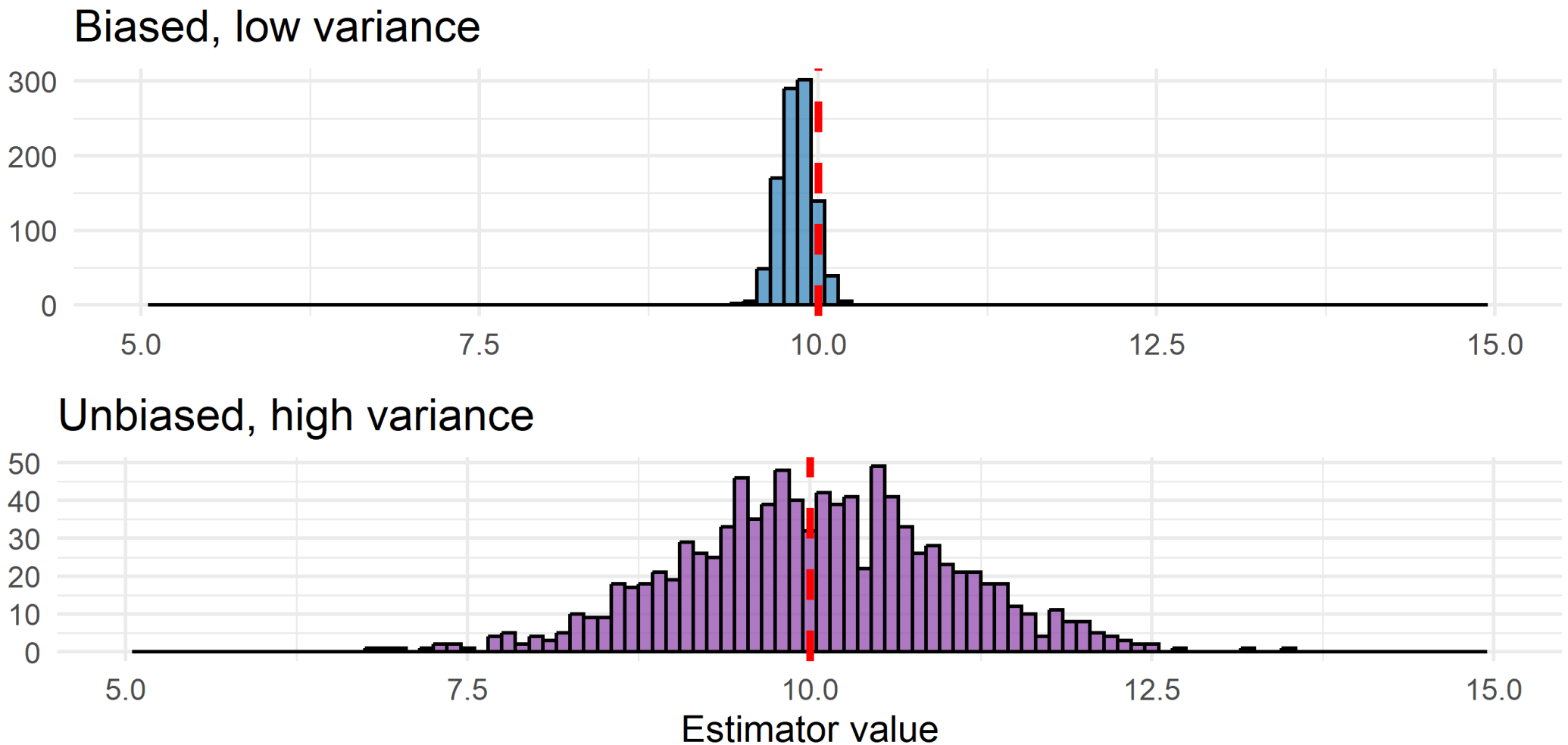
A lower MSE can mean a biased estimator

Nothing says the winner on MSE has to be unbiased. A method that is slightly off but very steady can beat one that is right on average but jumps around.

- MSE is one suggestion of which estimator is better — it weighs bias and variance against each other in a particular way.
- It is not a law. Whether you would rather be unbiased or have low variance depends on what the number is for.

Below: two estimators of the same parameter. Red dashed line = the true value.

Trading bias for variance: the picture



5 · The Central Limit Theorem

Centre and spread are settled. What is the shape?

From sampling distribution to the CLT

- We know the mean and variance of the sample mean. Can we say anything about its whole distribution?
- For the sample mean — yes!
- That’s the Central Limit Theorem, the most exciting theorem in statistics.

CLT: the setup

Take $X_1, \ldots, X_n$ i.i.d. with mean μ and standard deviation σ.

The theorem talks about three quantities, and they are the same thing rescaled:

	Definition	Centre	Spread
Sum	$S_n = \sum_{i=1}^{n} X_i$	$n\mu$	$\sqrt{n}\,\sigma$
Mean	$\bar{X}_n = \dfrac{S_n}{n}$	μ	$\dfrac{\sigma}{\sqrt{n}}$
Standardized	$Z_n = \dfrac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$	0	1

- Divide the sum by n and you get the mean: the centre drops from $n\mu$ to μ and the spread from $\sqrt{n}\sigma$ to $\sigma/\sqrt{n}$. Sums get *more* variable with n; averages get *less*.
- Subtract the centre and divide by the spread and you get Z_n — the same quantity with the units stripped off.

That $\dfrac{\sigma}{\sqrt{n}}$ has a name: the standard error of the mean. It is the standard deviation of $\bar{X}_n$, not of the data.

CLT: the theorem

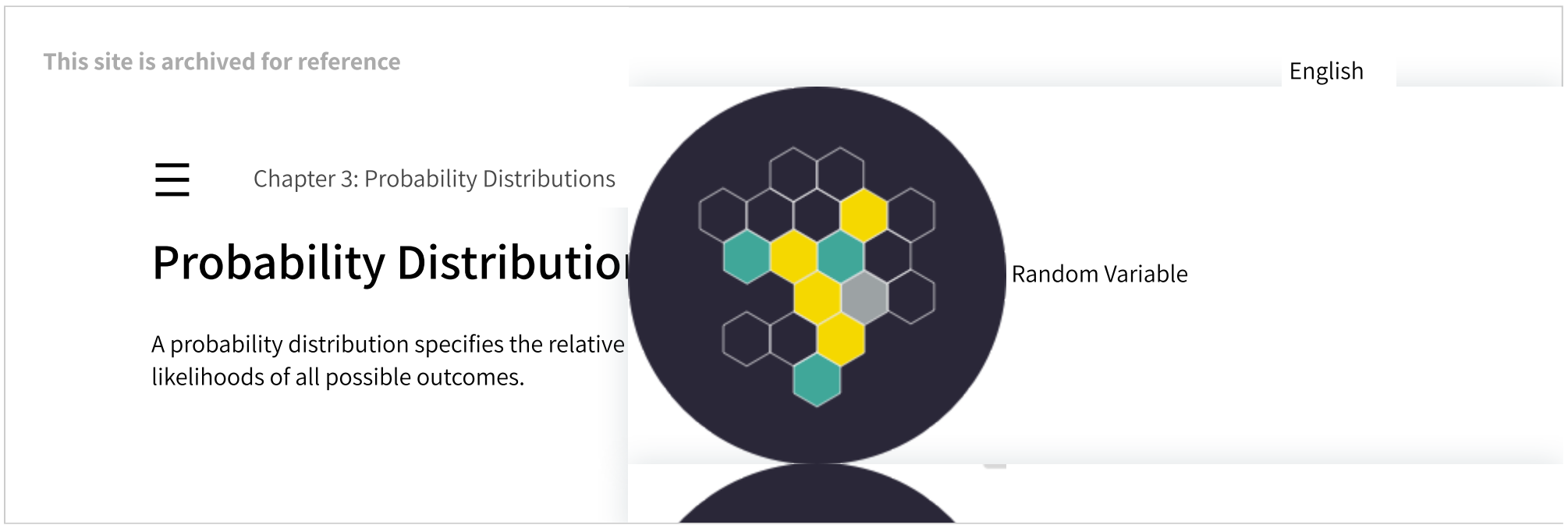
For large n:

$$S_n \dot{\sim} \mathcal{N}(n\mu,\, \sqrt{n}\,\sigma)\,, \quad \bar{X}_n \dot{\sim} \mathcal{N}\left(\mu,\, \frac{\sigma}{\sqrt{n}}\right), \quad Z_n \dot{\sim} \mathcal{N}(0,1)$$

- In large samples $\bar{X}_n$ is approximately normal, centered at μ, with st. dev. $\frac{\sigma}{\sqrt{n}}$.
- As n grows, the approximation improves and the spread shrinks with $\sqrt{n}$.

CLT: what drives the shape

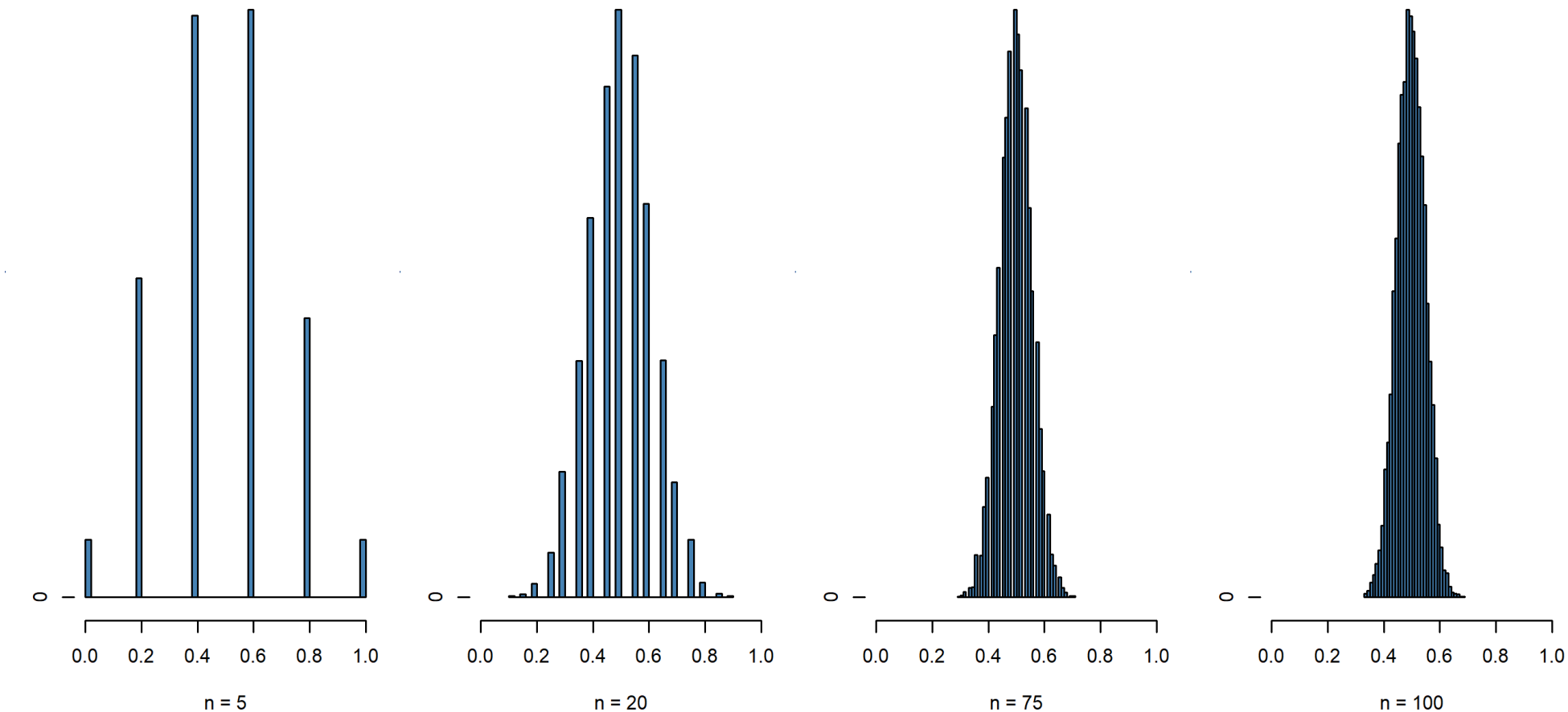
- The original distribution of X_i doesn't matter (but outliers slow convergence).
- Larger n → tighter distribution around the mean.
- Smaller σ → tighter distribution around the mean.



seeing-theory.brown.edu — pick any population shape and watch the sample means

CLT: simulate it (even for discrete data)

$X_i \sim \text{Bernoulli}(0.5)$. Distribution of $\bar{X}_n$ for growing n — it becomes normal and tighter:



- St. dev. of the mean: $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{\sqrt{p(1-p)}}{\sqrt{n}} = \frac{0.5}{\sqrt{n}}$.

What if I don't know σ?

Store sales are normal with mean 200 and unknown variance. Take $n = 80$ stores: $P\left(\bar{X} > 220\right)$?

- By the CLT, $\bar{X} \sim \mathcal{N}\left(200, \frac{\sigma}{\sqrt{80}}\right)$ — but σ is unknown.
- Use the sample standard deviation to estimate σ.
- That adds a little uncertainty, but in a large sample you estimate σ well.
- The distribution still converges to normal — you just need a few more observations.

Exercise 9 — A probability from the CLT

Monthly sales at your stores are normal with mean 200 (thousand pesos) and unknown variance. You have last month's figures for 80 stores in `store_sales`.

What is the probability that 80 stores average more than 220?

Four steps.

Exercise | Start Over | Show Hint | Run Code

```
1
```

→ More exam practice at the end of this deck

What you must be able to do this week

- Tell a parameter from a statistic — and, given a scenario in words, name the population, the sample, the parameter and the statistic. A parameter can be a mean, a proportion or a variance.
- Tell a random variable P_i from a realized value p_i, and state what makes a sample IID.
- Write an estimator as a function of the sample and compute a point estimate.
- Check an estimator for bias, compare two by variance/efficiency, and combine both via MSE = Var + Bias^2.
- Describe the sampling distribution of the mean and simulate it in R.
- State the CLT: $\bar{X}_n \approx \mathcal{N}(\mu, \sigma/\sqrt{n})$ — approximately normal, centered at μ, spread shrinking with $\sqrt{n}$.

Next week (W5): turn the sampling distribution into confidence intervals.

Practice for this week

Weekly exercise set: `W04_exercises.pdf`, with solutions in `W04_solutions.pdf`.

You can already solve exercises 1–12.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
1	Liga MX winning percentage	R as a calculator
3	Unemployment duration: probability for a sample mean	R as a calculator
6	Taquería and torta franchises: mean and variance of averages	R as a calculator
10	Standard error and sample size for spending per customer	R as a calculator
11	Sampling from ENIGH	R with the data

Appendix — optional theory refresher

Parameter or statistic?

For each one name all four — population, sample, parameter, statistic — and say which of them you actually have.

1. Spotify wants the share of songs listeners skip in the first 30 seconds. Its servers logged every play in Mexico last month: 24.1% were skipped.
2. A distillery must certify the mean alcohol content of a batch of 50,000 bottles. Testing opens the bottle, so 60 are tested and average 38.4%.
3. You want to know how long a DiDi takes to arrive in Polanco on a Friday night. You open the app at 20 random moments over a month; the average wait is 4.2 minutes.
4. A supplier wants to know how unevenly customers spend at Oxxo. From till records for 3,000 visits across 12 stores it computes a variance of 8,900 pesos^2.

Parameter or statistic? The answers

	Population	Sample	Parameter	Statistic
Spotify	every play in Mexico last month	none — the log covers all of them	π, the share skipped — it is 24.1%	none, nothing was estimated
Tequila	the 50,000 bottles in the batch	the 60 bottles opened	μ, mean alcohol content of all 50,000	$\bar{x} = 38.4\%$
DiDi	every Friday-night ride request in Polanco	the 20 moments you checked	μ, mean waiting time	$\bar{x} = 4.2$ min
Oxxo	every Oxxo visit	the 3,000 recorded visits	σ^2, variance of spending	$s^2 = 8{,}900$ pesos^2

- Spotify: being computed by a machine from millions of rows does not make a number a statistic. Ask for *next* month's rate instead and everything changes — that population has not happened, so it can only be estimated.
- Tequila: μ exists and will never be measured. Measuring it means destroying everything you were going to sell.
- DiDi: $\bar{x}$ estimates μ only if the 20 moments were spread across the night. Twenty checks all made at 9 pm estimate the 9 pm wait.
- Oxxo: a spread is a parameter exactly as much as a mean is. What makes a number a statistic is where it was computed, never what kind of number it is.

→ Four more to work through · ← Back to Statistical language

Name the four pieces

What is the population, the sample, the parameter and the statistic here?

- You want to know the probability that a user who got a match on Tinder goes on a date with that person. You survey 1000 users and ask, for each match they got, whether they went on a date. You then calculate the share of matches that ended in a date for those users.
- You want to know whether Starbucks baristas are faster than Cielito Querido baristas. You go to 10 Starbucks and 10 Cielito Querido and measure how long it takes to make a coffee. You then calculate the average time in each chain.
- You want to know the average age of people who go to the gym. You go to a gym and ask 100 people their age. You then calculate the average age of those people.
- You want to know the variance of internet speed in Mexico City. You visit 500 households and calculate the variance of their internet speed.

Name the four pieces: the answers

	Population	Sample	Parameter	Statistic
Tinder	every match of every user	the matches reported by the 1000 users surveyed	true share of matches that end in a date	share of dates among the reported matches
Baristas	every coffee made in each chain	the coffees timed in the 20 shops visited	mean preparation time in each chain	mean preparation time in the shops visited
Gym	everyone who goes to the gym	the 100 people asked	mean age of all gym-goers	mean age of the 100
Internet	all households in Mexico City	the 500 households visited	variance of speed across all households	variance of speed across the 500

- The parameter is always a number about the left-hand column; the statistic is the same recipe applied to the right-hand one.
- Note the last row: a variance is a parameter too. Anything that describes the population qualifies — a mean, a proportion, a spread, a difference between two groups.

← Back to Statistical language

Sampling distribution of the standard deviation

- Sample means are normal in large samples. What about the sample standard deviation?
- If we take many samples and compute each one's SD, what does *that* distribution look like?

- If X_i is normal, the sample variance is related to the chi-squared distribution.
- This is what we'll use to build confidence intervals for the variance.

From Normal to Chi-Square

Start with the standard normal $Z \sim N(0, 1)$ and square it. The distribution of $X = Z^2$ is called chi-squared with one degree of freedom, $\chi^2(1)$.

That is the definition. The rest you can find out yourself:

- What shape does it have, and what values can it take?
- What are its mean and its variance?
- If $X = Z^2$, how does $P(X > 1)$ relate to a probability about Z?

Exercise 14 — Square a normal

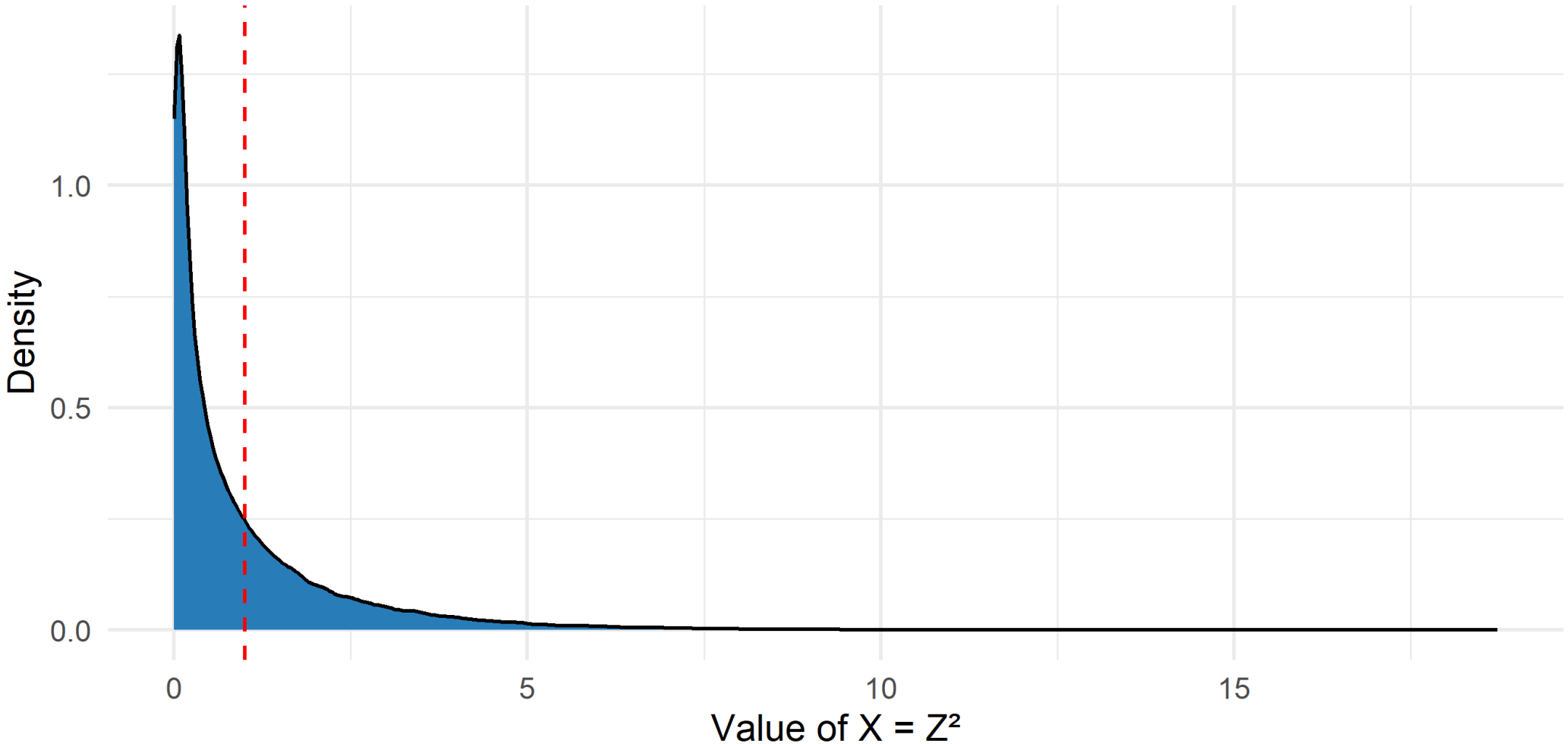
1. Draw 100,000 standard normals into `z` and square them into `x`.
2. Plot the distribution of `x`, and compute `mean(x)` and `var(x)`.
3. Compute `mean(x > 1)` and `mean(abs(z) > 1)`. Are they equal? Say in one line why.

Exercise | Start Over | Show Hint | Run Code

```
1  # Your code here
```

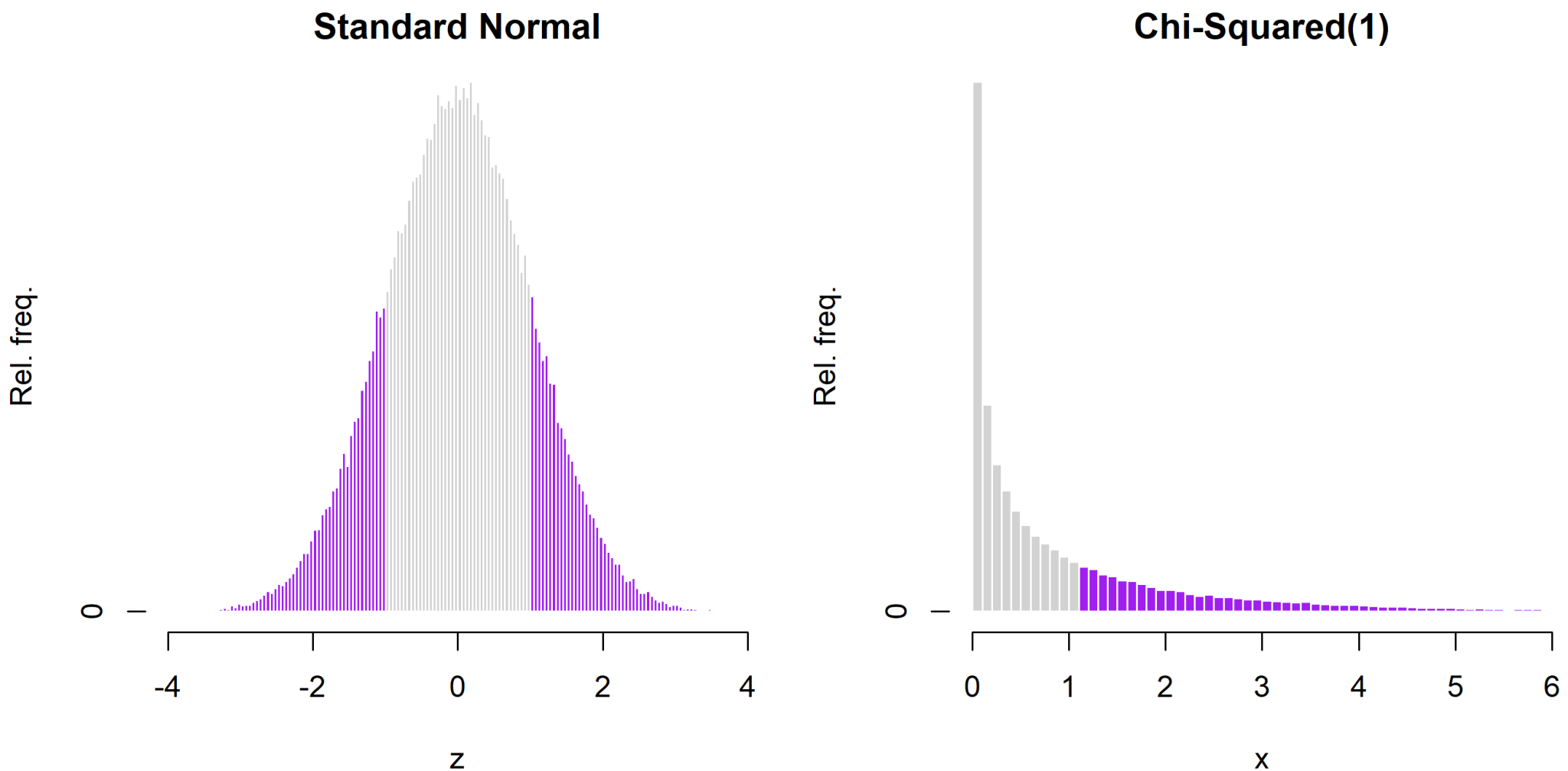
Chi-Square: the picture

- $E[\chi^2(1)] = Var(Z) = 1; Var(\chi^2(1)) = 2.$



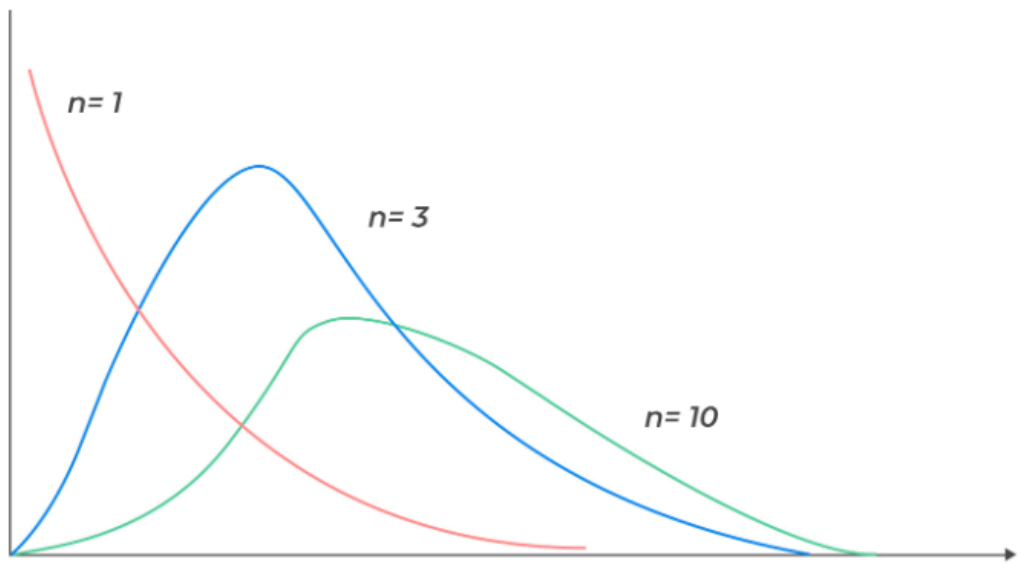
Visualizing the connection

Shaded area = $P(X = Z^2 > 1)$ where $X \sim \chi^2(1), Z \sim N(0,1)$ — same area in both panels:



Chi-Square & the sum of normals

- Sum of n i.i.d. squared standard normals is $\chi^2(n)$: $\sum_n Z^2 \sim \chi^2(n)$.
- $E[\chi^2(n)] = n; Var(\chi^2(n)) = 2n.$



- Why does the shape approach normal for large n? Because it's a sum of random variables — the CLT again.

Where $Var(\bar{X}) = \sigma^2/n$ comes from

Two rules do all the work: a constant comes out of a variance squared, and the variance of a sum is the sum of the variances plus every covariance.

$$Var(\bar{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2} Var\left(\sum_{i=1}^{n} X_i\right)$$

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + \underbrace{\sum_{i \neq j} Cov(X_i, X_j)}_{\text{every pair, twice}}$$

Now use independence. If the listings are drawn independently, every $Cov(X_i, X_j)$ with $i \neq j$ is zero, and the whole second term disappears:

$$Var\left(\sum X_i\right) = \sum_{i=1}^{n} \sigma^2 = n\sigma^2$$

$$Var(\bar{X}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

So the $\sqrt{n}$ law is not free. It is bought entirely with the independence assumption — see what happens without it.

Where $\frac{10\sigma^2}{16}$ comes from

Same two rules, applied to $\hat{\theta}_1 = \frac{3X_1+X_2}{4}$.

$$Var(\hat{\theta}_1) = Var\left(\frac{3X_1 + X_2}{4}\right) = \frac{1}{16}\,Var(3X_1 + X_2)$$

$$= \frac{1}{16}\Big[\,9\,Var(X_1) + Var(X_2) + 2 \cdot 3 \cdot Cov(X_1, X_2)\,\Big]$$

Independent draws, so the covariance is zero and each variance is σ^2:

$$= \frac{9\sigma^2 + \sigma^2}{16} = \frac{10\sigma^2}{16}$$

The 9 is where the damage is done: tripling the weight on X_1 multiplies its contribution by nine, and nothing else is there to offset it.

← Back to Variance: worked examples

Where MSE = Var + Bias^2 comes from

Add and subtract $E(\hat{\theta})$ inside the square, then expand.

$$
\begin{aligned}
MSE(\hat{\theta}) &= E[(\hat{\theta} - \theta)^2] = E\big[\big(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta\big)^2\big] \\
&= E[(\hat{\theta} - E(\hat{\theta}))^2] + 2\underbrace{E[\hat{\theta} - E(\hat{\theta})]}_{=0}(E(\hat{\theta}) - \theta) + (E(\hat{\theta}) - \theta)^2 \\
&= \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2
\end{aligned}
$$

The trick is the middle line: the cross term vanishes because an estimator's average deviation from its own average is zero, by definition.

← Back to MSE = Variance + Bias^2

What if the draws are not independent?

Start with two. The variance of a sum is not just the sum of the variances:

$$Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2\,Cov(X_1, X_2)$$

$$Var(X_1 - X_2) = Var(X_1) + Var(X_2) - 2\,Cov(X_1, X_2)$$

With n of them the same thing happens for every pair:

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) \;+\; \sum_{i \neq j} Cov(X_i, X_j)$$

so the sample mean picks them up too:

$$Var(\bar{X}) = \frac{\sigma^2}{n} \;+\; \frac{1}{n^2} \sum_{i \neq j} Cov(X_i, X_j)$$

Positive covariance makes it bigger than σ^2/n; negative covariance makes it smaller. Independence sets every one of those terms to zero.

← Back to Variance: worked examples

Trading bias for variance

Estimating customer income; the true value is $\theta = 10$. Two candidates:

- A — biased by -0.2, but almost no noise (sd 0.1)
- B — unbiased, but noisy (sd 1)

Unbiasedness alone says take B. Which one would you actually report? Produce the numbers and the pictures before you decide.

Exercise 13 — Put a number on the trade-off

Simulate 5,000 draws of each estimator. Then, for both:

1. Compute the bias, the variance, and the MSE two ways — straight from the definition `mean((est - theta)^2)`, and from the decomposition $Var + Bias^2$.
2. Draw the two sampling distributions side by side on a common x-axis, with the true value marked.
3. Say which one you report, and why.

Exercise | Start Over | Show Hint | Run Code

```
1 theta <- 10
2 A <- rnorm(5000, mean = 9.8, sd = 0.1)
3 B <- rnorm(5000, mean = 10,  sd = 1)
4
5 # bias, variance, MSE (both routes) for each; then the two histograms
```

Exercise 7 — Theory vs. simulation

The CLT claims $\bar{X}_{200} \dot{\sim} \mathcal{N}(\mu, \sigma/\sqrt{30})$. Use it, then check it.

1. Theory, no data: with `mu <- mean(population)` and `se <- sd(population)/sqrt(200)`, compute the probability that a sample of 200 returns a mean above 1,250 pesos.
2. Simulation: build 5,000 sample means and compute the share of them above 1,250.
3. How close are the two answers?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Does the shape of the population matter?

`population` is strongly skewed, and its sample means still came out symmetric. Maybe real prices are a lucky case.

Two more populations are loaded:

- `pop_bimodal` — two clumps of listings: cheap rooms around 600 pesos, design apartments around 2,500.
- `pop_binary` — a 0/1 variable: is the listing a superhost?

Neither is remotely bell-shaped. Guess what the histogram of their sample means looks like.

Exercise 8 — Start from something not normal

Put three pictures side by side with `par(mfrow = c(1, 3))`:

1. `hist(pop_bimodal)` — the population itself.
2. the histogram of 2,000 sample means with `n = 5`.
3. the same with `n = 50`.

Then repeat the whole thing starting from `pop_binary`.

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exercise 9 — Break it: more replications

A classmate says: *“my standard error is too big — I’ll just run more replications.”*

Test the claim. Hold `n = 200` fixed and build `means` with 100, 1,000 and 10,000 replications. Report `sd(means)` each time, and say what *does* change.

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exercise 10 — The whole week, one question

Back to the Airbnb portfolio. Your investor asks:

> *“You surveyed 25 listings and got an average nightly price. How wrong could that number be?”*

Answer with a number, a picture, and one sentence.

Exercise | Start Over | Show Hint | Run Code

```
# 1. A number: how far from the truth does an estimate from n = 25 typically land?

# 2. A picture: the sampling distribution for n = 25, with the truth marked
#    and the range that holds 90% of the estimates.

# 3. One sentence: what do you tell the investor?
```

Normal distribution: combining variables

A customer who opens the DiDi app may call a car. X = calls in Cancún (Bernoulli 0.4), Y = in Puerto Vallarta (Bernoulli 0.6). Sample: 100 from Cancún, 80 from PV. P(more than 100 people call a car)?

Reminders (for normal X):

$$X + c \sim \mathcal{N}(\mu + c, \sigma), \qquad cX \sim \mathcal{N}(c\mu, |c|\sigma)$$

$$X + Y \sim \mathcal{N}\left(\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2}\right)$$

Appendix: building the sampling distribution by simulation

The same facts as the formulas above, measured instead of derived.

Exercise 3 — Once, then two thousand times

1. Draw one sample of `n = 200` from `population`, store it in `s`, compute `mean(s)`.
2. Now do that 2,000 times. Collect the 2,000 means in `means`.
3. `hist(means)`, then mark the parameter on the picture with `abline(v = mean(population), col = "red", lwd = 3)`.

Look at the picture before moving on: where is it centred, and how wide is it compared with the prices themselves (`hist(population)`)?

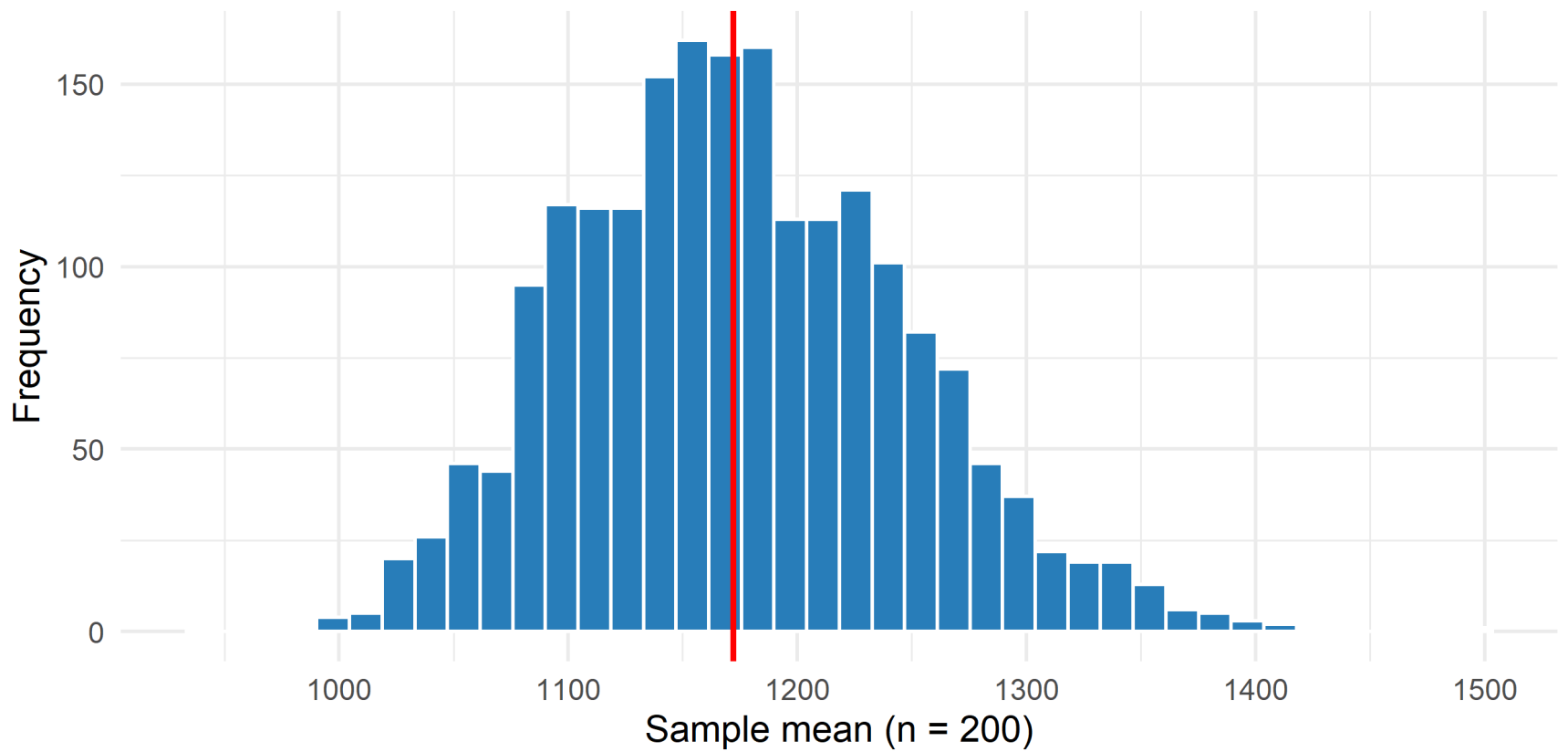
Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

What you just built: the sampling distribution

The distribution of an estimator across all the samples it could have been computed from.

```
set.seed(1)
means <- replicate(2000, mean(sample(population, 200)))   # population = the 21,184 prices
ggplot(data.frame(means), aes(means)) +
  geom_histogram(bins = 40, fill = "#2980b9", color = "white") +
  geom_vline(xintercept = mean(population), color = "red", linewidth = 1.2) +
  labs(x = "Sample mean (n = 200)", y = "Frequency") + th
```



The means cluster around the true mean (red) — the estimator has a *distribution*.

How wide is that histogram?

Individual prices are spread out: `sd(population)` is about 1,021 pesos.

The sample means sat in a visibly narrower band. How much narrower?

Write down a guess for `sd(means)` at `n = 200` before you compute it.

Exercise 4 — Guess the spread, then measure it

Your guess for `sd(means)`: _____

Now rebuild `means` and compute three numbers:

1. `sd(population)` — the spread of individual prices.
2. `sd(means)` — the spread of the sample means.
3. the ratio `sd(population) / sd(means)`.

That third number is one you recognise. Which number is it?

Exercise | Start Over | Show Hint | Run Code

```
1 # Your code here
```

Exercise 5 — Break it: change n

Here is working code. Change one thing — the sample size — and report what happens to the centre and to the spread.

Exercise | Start Over | Show Hint | Run Code

```r
n <- 5
means <- replicate(2000, mean(sample(population, n)))
c(centre = mean(means), spread = sd(means))

# Run it again with n = 50, n = 200, n = 800.
# Which of the two numbers moves? By how much when n goes from 5 to 800?
```

Exercise 6 — As n grows…

As the sample size n grows, the sampling distribution of the mean gets ___

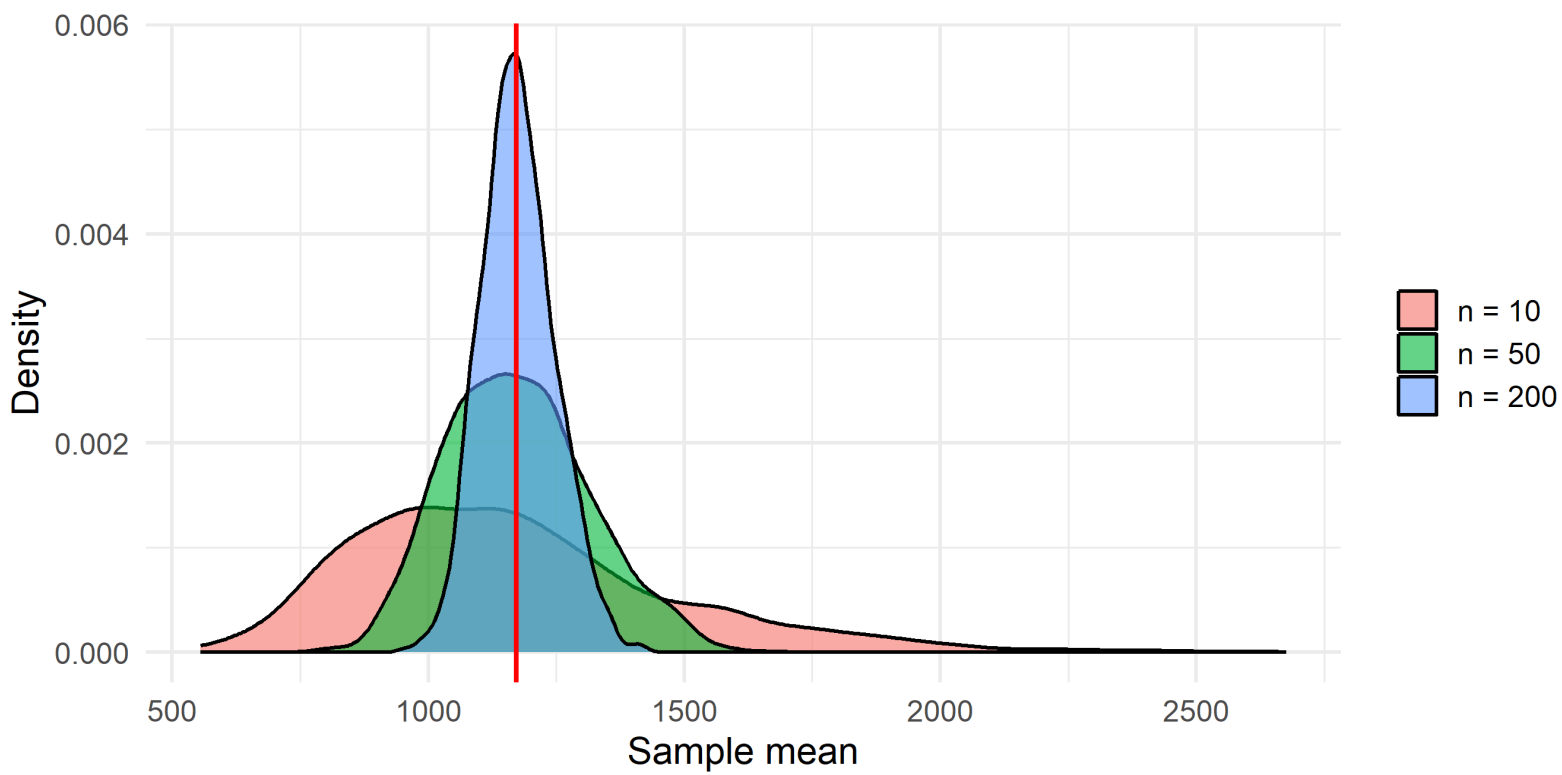
- narrower (smaller spread)?
- wider?
- more skewed?
- shifted away from the true mean?

Narrower — and by a specific amount: the spread you measured is $sd(\text{population})/\sqrt{n}$, which shrinks as n grows while the centre stays on the parameter.

The spread shrinks with $\sqrt{n}$

Same population, three sample sizes — centred in the same place, tighter every time:

```
set.seed(7)
pop <- population                                  # the 21,184 real prices
df <- do.call(rbind, lapply(c(10, 50, 200), function(n)
  data.frame(mean = replicate(1500, mean(sample(pop, n))), n = paste0("n = ", n))))
df$n <- factor(df$n, levels = c("n = 10", "n = 50", "n = 200"))
ggplot(df, aes(mean, fill = n)) +
  geom_density(alpha = 0.6) +
  geom_vline(xintercept = mean(pop), color = "red", linewidth = 1) +
  labs(x = "Sample mean", y = "Density", fill = NULL) + th
```



Exercise 11 — Which ones are unbiased?

The code is written for you: it draws 5,000 samples of `n = 10`, computes all three estimators in each one, and reports two rows — the bias of each estimator (its average minus the parameter) and how spread out each one is.

Press Run, then answer from the output:

1. Which of the three are unbiased?
2. Among those, which one would you rather use — and why?

Exercise | Start Over | Show Hint | Run Code

```
mu <- mean(population)

est <- replicate(5000, {
  x <- sample(population, 10)
  c(theta1 = x[1],
    theta2 = (3 * x[1] + x[2]) / 5,
    theta3 = mean(x))
})

round(rbind(bias   = rowMeans(est) - mu,
            spread = apply(est, 1, sd)), 1)
```

Exercise 12 — Which unbiased estimator wins?

Simulate 1,000 samples of four draws from $\mathcal{N}(10, 10)$, compute both estimators in each sample, then produce the evidence: their averages, their standard deviations, and the two histograms side by side.

Exercise | Start Over | Show Hint | Run Code

```
1  # one sample of four draws:   x <- rnorm(4, mean = 10, sd = 10)
2  # build A and B from x, repeat 1,000 times, then compare and plot
```

Exam practice

These problems mirror the weekly exercise set and the style of the exam — work each one in the code cell, then press Solution to reveal the full worked answer.

← Back to the exam spotlight

Exam practice 1 — Liga MX winning percentages

Liga MX teams play 34 games per season. A team's final winning percentage is its number of wins divided by 34. Assume each match is an i.i.d. Bernoulli trial with the team's "true" win probability. [l1]

a. If Club América has win probability $\pi_A = 0.58$, what is the asymptotic distribution of América's winning percentage W_A?

b. If América has $\pi_A = 0.58$ and Pumas has $\pi_B = 0.52$, what is the asymptotic distribution of $W_A - W_B$, assuming independence?

c. Based on (b), what is $P(W_B > W_A)$?

Exercise | Start Over | Show Hint | Run Code

```
n   <- 34
piA <- 0.58; piB <- 0.52

# (a) the mean and the variance of America's winning percentage
# (b) the mean and the variance of the difference W_A - W_B
# (c) the probability that that difference comes out negative
```

Exam practice 2 — Factory injuries (Poisson)

In a factory in Monterrey, the number of workplace injuries is recorded weekly for 240 weeks. The average number of injuries per week is 0.6, with $X_t \sim \text{Poisson}(\lambda)$. [I2]

a. Provide an asymptotic 90% confidence interval for λ.

Exercise | Start Over | Show Hint | Run Code

```
lambda_hat <- 0.6
n          <- 240

# (a) the standard error of lambda-hat, then the 90% interval around it
```

Exam practice 3 — Unemployment duration: concepts

A random sample of 500 unemployed workers in Guadalajara is surveyed about unemployment duration in weeks, X. By the CLT, $\bar{X}$ is approximately normal. [I8]

a. If $P(\bar{X} > 20) = 0.5$, what is $E(X)$?

b. If $P(\bar{X} > 20) = 0.4$, what can be said about $E(X)$?

Answer them by finding the mean that produces each probability. Take $\sigma_X = 10.5$, so $SE = 10.5/\sqrt{500}$.

Exercise | Start Over | Show Hint | Run Code

```
1  se <- 10.5 / sqrt(500)
2
3  # (a) the mean that makes P(xbar > 20) equal to 0.50
4  # (b) the mean that makes P(xbar > 20) equal to 0.40
```

📝 Exam practice 3 — Unemployment duration: calculations

Same setup: $n = 500$ workers, $\bar{X}$ approximately normal by the CLT. [I8]

c. If $E(X) = 19.5$ and $\sigma_X = 10.5$, what is $P(\bar{X} > 20)$?

d. How would the answer to (c) change if $n = 1600$ instead of $n = 500$?

Exercise | Start Over | Show Hint | Run Code

```
mu    <- 19.5
sigma <- 10.5

# (c) the probability that the sample mean exceeds 20 when n = 500
# (d) the same probability when n = 1600
```

Exam practice 4 — Cereal box weights

A cereal company in Guadalajara sells 500g boxes. Box weights follow $X \sim N(500, 9)$ (variance 9, so $\sigma = 3$). Boxes weighing less than 496g must be repackaged. [I9]

a. What is the probability that a box is repackaged?

b. For 15,000 boxes, what is the approximate (normal) distribution of the number repackaged?

Exercise | Start Over | Show Hint | Run Code

```
1  n <- 15000
2
3  # (a) the probability that one box comes out under 496g
4  # (b) the mean and variance of the NUMBER of repackaged boxes out of 15,000
```

c. Profits per box are P_1 if at least 496g and P_2 if less. For 15,000 boxes, what is the approximate distribution of total profits?

Total profit is $\Pi = (15000 - Y)P_1 + YP_2 = 15000\,P_1 - Y(P_1 - P_2)$. With $a = P_1 - P_2$, a linear function of the (approximately normal) Y is normal:

$$\Pi \dot{\sim} N\big(15000\,P_1 - 1368.2\,a,\;\; 1243.4\,a^2\big)$$

Exam practice 5 — Online marketplace visitors

The number of visitors X to a Mexican online marketplace during one minute is $\text{Poisson}(\lambda = 150)$. [l10]

a. Thinking of X as the sum of 60 i.i.d. Poisson(2.5) random variables, what is the asymptotic normal distribution associated with X?

b. Provide an approximate 90% probability interval for X using the asymptotic distribution.

c. Calculate the exact probability (based on $\text{Poisson}(150)$) that X is within the interval from (b).

Exercise | Start Over | Show Hint | Run Code

```
1  lambda <- 150
2
3  # (a) the mean and variance of X, seen as a SUM of 60 iid Poisson(2.5)
4  # (b) an interval around that mean holding about 90% of the distribution
5  # (c) the exact Poisson probability of landing inside that interval
```

Bonus: why is $S = \sum_{i=1}^{n} X_i$ an unbiased estimator of $n\mu_X$?

$E[S] = E[\sum_{i=1}^{n} X_i] = \sum_{i=1}^{n} E[X_i] = n\mu_X$ — the expected value of S equals the parameter, which is exactly the definition of unbiasedness.

Exam practice 6 — Franchise revenues

You own 60 taquería franchises and 30 torta franchises. Monthly revenue (thousands of pesos) for each franchise is i.i.d. with: [I14]

- Taquerías: $\mu = 22, \sigma = 5$
- Tortas: $\mu = 17, \sigma = 4$

a. What is the approximate distribution of the sample average of revenues for the 30 torta franchises?

b. What is the approximate distribution of the sample average of revenues across all 90 franchises?

Exercise | Start Over | Show Hint | Run Code

```
1 # (a) the mean and variance of the average revenue of the 30 torta franchises
2 # (b) the mean and variance of the average revenue across all 90 franchises
```